BioPlast 'IA -AI for bio-based plasticizers

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INTRODUCTION

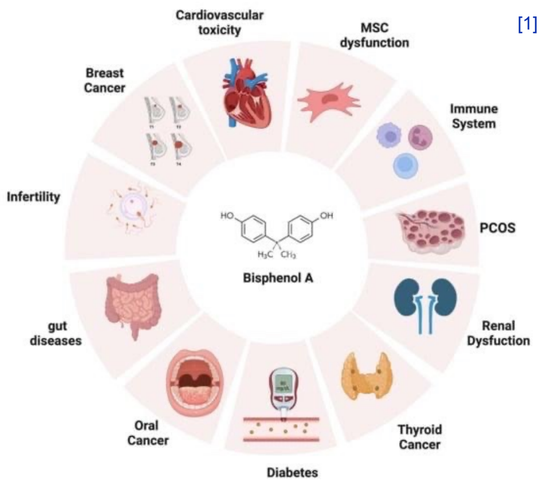
Plasticizers represent **10 to 40% of the formula** of plastics.

Phthalates are the **most commonly used** plasticisers. They are known to be **toxic to the environment and human health**.

BioPlast'IA aims to create **an artificial intelligence** that finds **sustainable plasticisers** while preserving the properties of the polymer.

The aim is to produce a plastic incorporating a **bioplasticizer**.

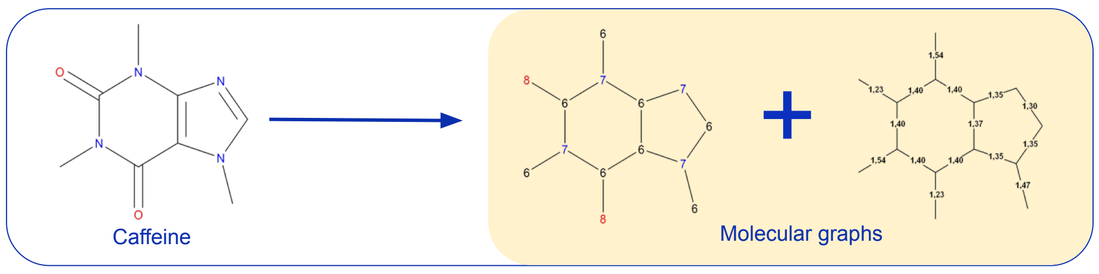
An **example case** will be used for the reasoning that it is **not a plasticizing molecule**.



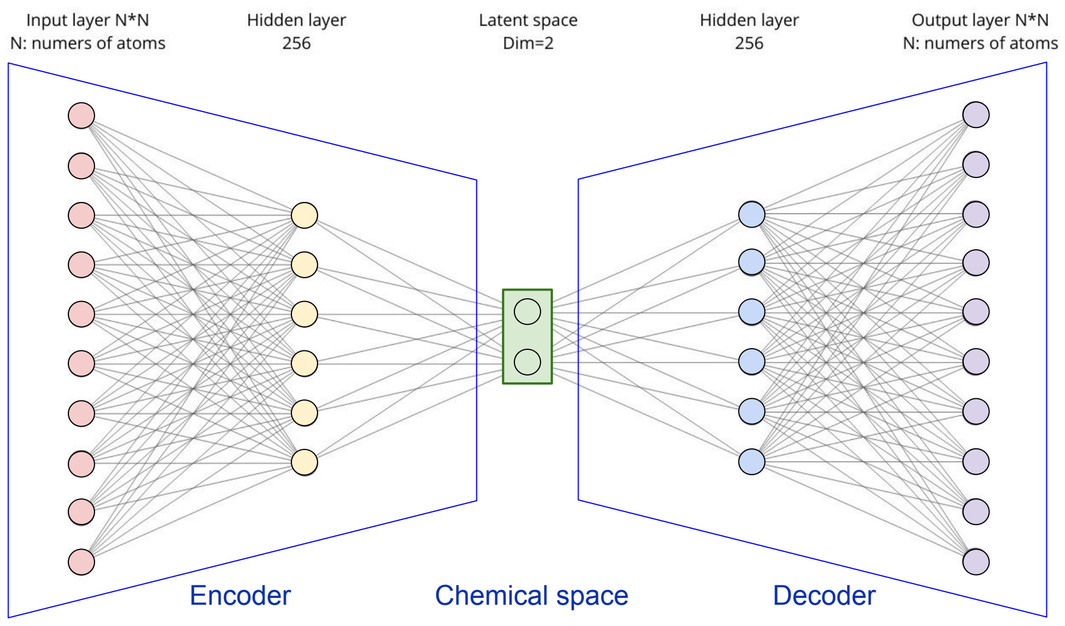
[1]

AUTOENCODER

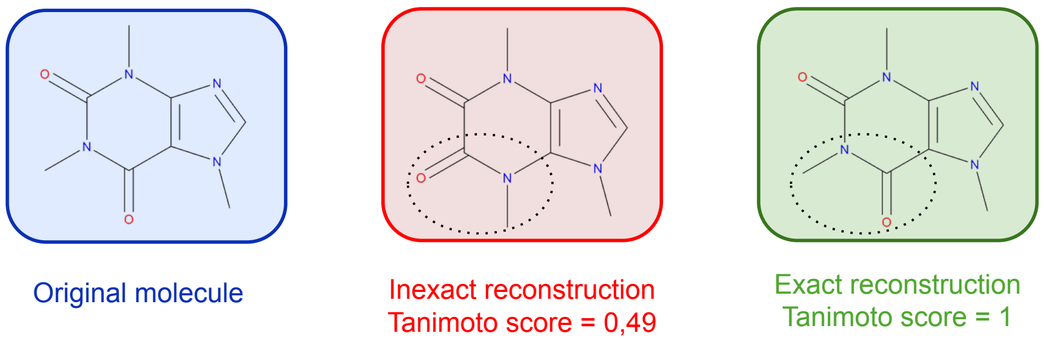
Conversion of **molecules** to **molecular graphs**



Architecture of the neural network



We **mapped** molecules into a **chemical space** (named latent space) to analyze their plasticizing properties. By using molecular descriptors and machine learning models, we **identified candidate molecules** close to **known plasticizers** in the chemical space.



The average **similarity Tanimoto score**[3] for reconstruction is **95%**

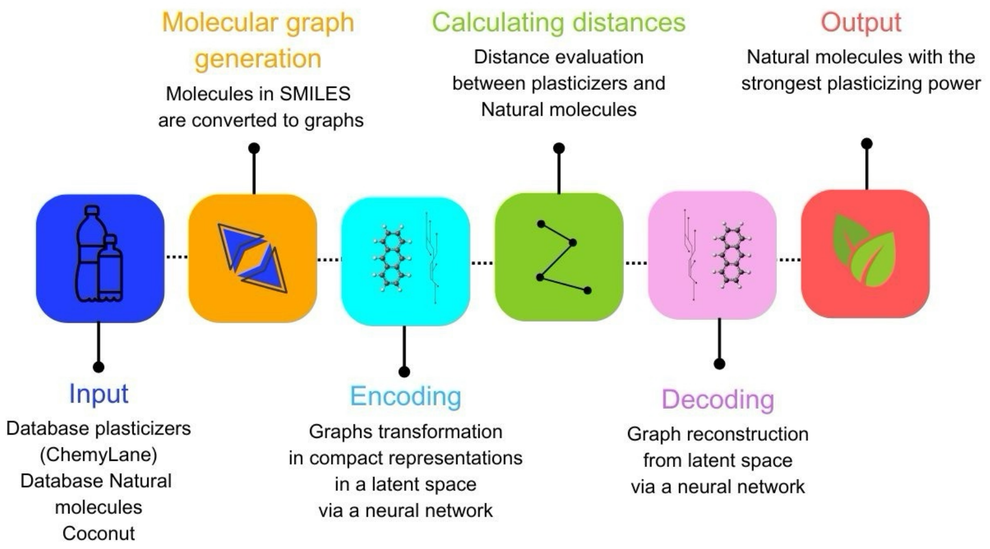
Hyperparameters of the model: Loss: BCE, Optimizer: Adam, Learning rate: 0,01, Epoch: 500, Batch normalization

CONCLUSION AND PERSPECTIVES

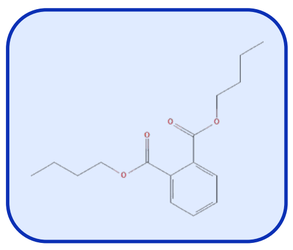
We successfully identified **two new molecules** with plasticizing properties that can replace **toxic plasticizers** currently in use. These molecules are widely available and could be **supplied to factories** for **industrial application.**

Looking ahead, we are working on **automating and optimizing** the search for suitable plasticizer candidates. Currently, the validation process is done manually, but our goal is to develop a system that **streamlines and accelerates** this selection process **using artificial intelligence.**

METHOD



Training dataset	Chemy Lane	95 plasticizers molecules extracted
Prediction dataset	COCONUT[2]	700 000 natural molecules (Updated daily)



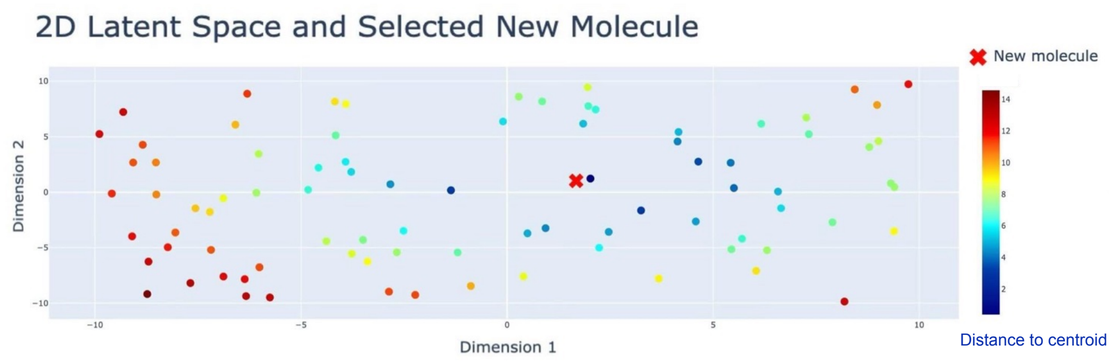
Example of a molecule from the training dataset
Dibutyl phtalate

A **plasticizer** is a substance added to polymers to increase their flexibility, workability, and elasticity by **reducing intermolecular forces**.

The **training dataset** contains **known plasticizers** such as phthalates, adipates, citrates, epoxidized oils and polymeric plasticizers

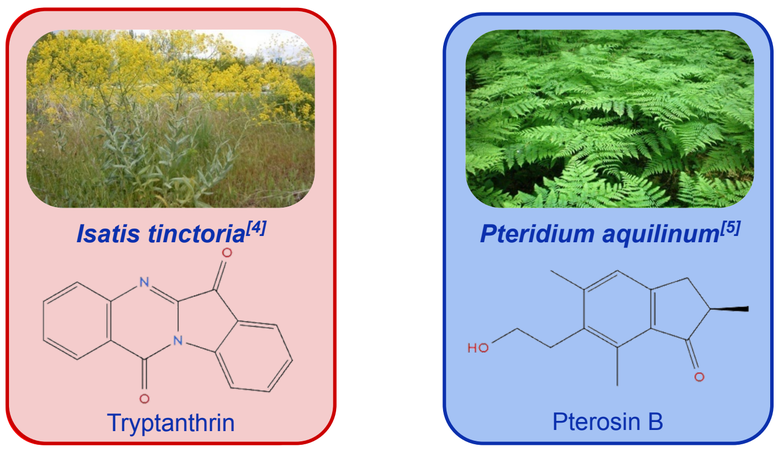
RESULTS

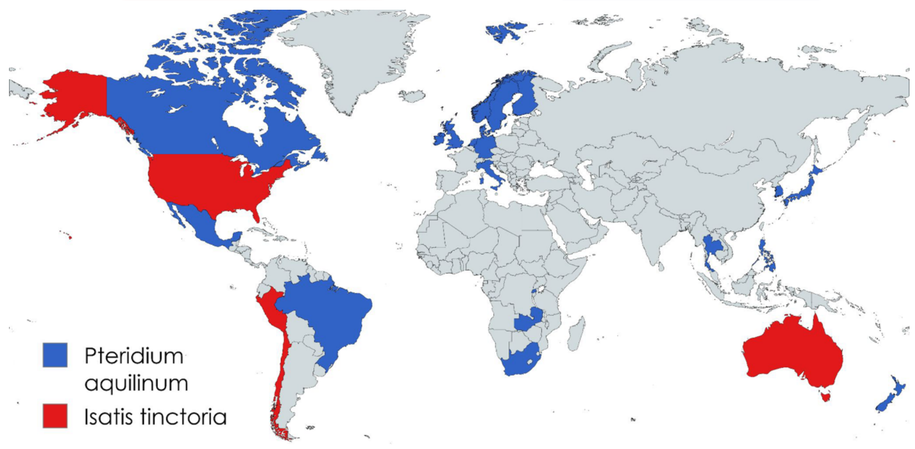
Latent space



We identified two molecules with expected **strong plasticizing properties**, positioned in the latent space near **known plasticizers**. These molecules were selected based on their **similarity** to existing plasticizers and their **availability**. Their potential for **industrial application** was confirmed, making them viable **alternatives** to toxic plasticizers. The **distance-to-centroid** analysis further supported their relevance within the **chemical space** of effective plasticizers.

Worldwide invasive plants





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